Recorder Overview

Recorder is a part of Autofill system which will be built to speed up repetitive applying to scholarship programs for students and educational agencies, recorder's goal is to collect data for future Autofill system. The Recorder captures a human operator's portal interactions section by section and stores structured JSON artifacts. These recordings provide training and ingestion data for the Portal Autofill system while keeping the operator in control of login, verification, navigation, data entry, and submission decisions.

Recording scope

The recorder supports four session categories:

1- AUTH - authentication and login flows
2- Registration - account creation and signup flows
3- Application - scholarship, program, degree, or major selection
4- Data Entry - application-form sections and field interactions

AUTH, Registration, and Data Entry sessions are organized by portal family so that recordings can be reused across universities that share the same platform. Application sessions remain university-specific because program and scholarship choices may differ within the same portal family.

Operator workflow

- Start the recorder from the project directory. The application loads the available universities from dataset.csv.
- Choose a session category.
- Select a local Chrome profile. The recorder copies it to a temporary profile directory so existing login state can be used without locking the live profile.
- Review the selected category, university, portal URL, and Chrome profile, then press Enter to start.
- Human completes the target flow manually in Chrome. Handle passwords, two-factor authentication, CAPTCHAs, navigation, form entry, and any portal-specific decisions as normal.
- Return to the terminal and press Enter when the target session is complete. Ctrl+C or the configured duration limit can also stop the recording.
- The recorder saves the action log and per-step form schemas, closes the browser session, and removes the temporary Chrome profile.

What the recorder captures

5- Input and change events for text fields, text areas, selects, checkboxes, and related controls
6- Clicks on buttons, links, submit controls, and button-like elements
7- Form submissions and URL navigation between pages
8- Initial page schemas, schemas after navigation, and schemas for dynamically revealed panels or newly added fields
9- Field metadata such as element type, name, ID, label, attributes, visibility, XPath, current URL, and detected step
10- Timestamps, the active step, and the university/session identifiers needed to organize the recording

Output structure

Each recording is written below the configured sessions directory using the category, organization key, university name, and short session ID. A typical session contains:

11- actions.json - session metadata and the chronological interaction log
12- step_00_initial_load.json - the initial form schema
13- step_XX_<step>.json - additional schemas captured after navigation, panel reveals, or DOM growth

Saved sessions can be processed with the ingest command to populate field mappings and the recorder's knowledge base.